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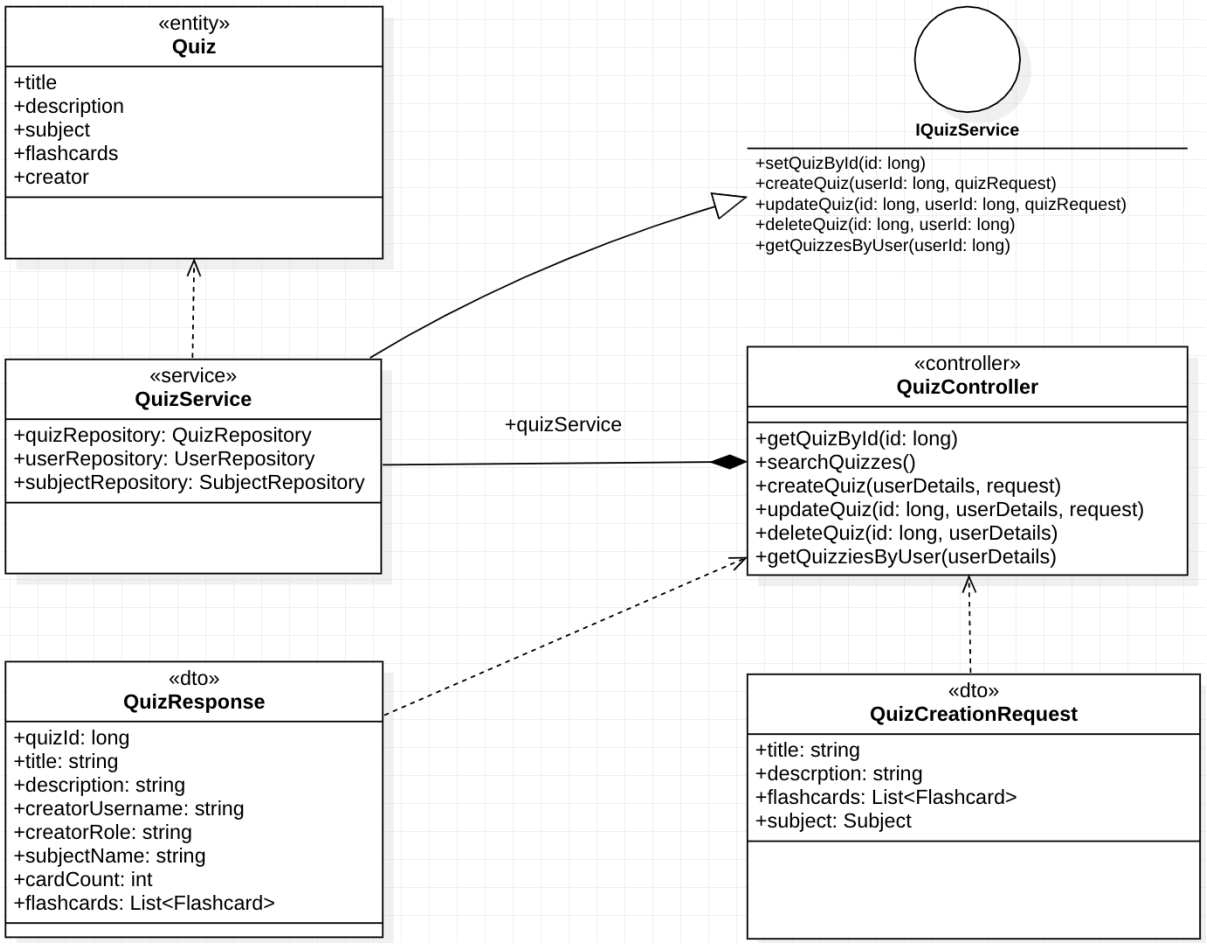
Online-flashcards Class Diagram

1. Class Diagram

The following class diagram depicts the class structure of a singular Spring Boot resource/endpoint in the application. The application follows the MVC pattern, but as the view part is implemented with React, it was not taken into account in the diagram. The application itself contains multiple of these resources, which are all very similar, so we opted to model just one of them:



2. Class Descriptions

- **Quiz:** JPA Entity class representing a quiz in the database. Contains the quiz's core data: title, description, subject, associated flashcards, and the user who created it.

- **IQuizService:** Interface defining the contract for quiz-related operations. Declares methods for creating, reading, updating, and deleting quizzes.

- **QuizService:** Service class that implements IQuizService. Contains the logic for quiz operations and coordinates between the quiz, user, and subject repositories.

- **QuizController:** REST controller that exposes quiz-related API endpoints to the client. Receives HTTP requests, delegates processing to QuizService, and returns appropriate responses.

- **QuizResponse:** Data Transfer Object used to shape outgoing quiz data sent back to the client. Includes flattened fields like creator username, role, subject name, and card count rather than exposing raw entity relationships.

- **QuizCreationRequest:** Data Transfer Object used to capture incoming data when a client creates or updates a quiz. Carries the title, description, flashcards, and subject provided by the user.